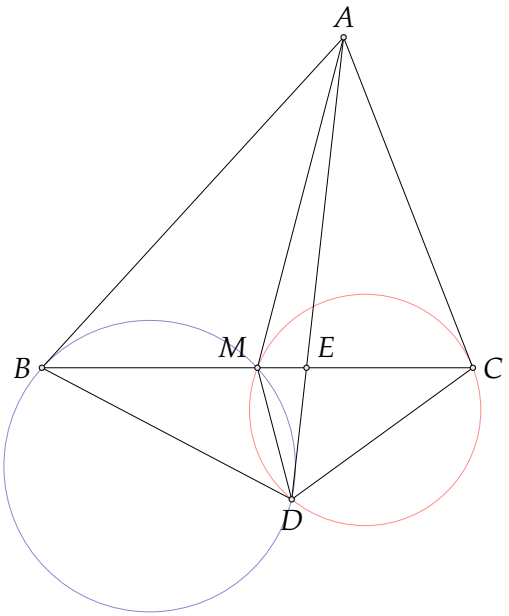
BMO1 2025 (November 2025), Problem 4

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The problem

Let ABC be an acute-angled triangle with $AB > AC$. Let M be the midpoint of BC. The circle passing through M that is tangent to AB at B and the circle passing through M that is tangent to AC at C intersect again at D. Prove that $MA \times MD = MB \times MC$.



The point E is where AD meets BC, and $E \neq M$ unless $AB = AC$.

Script A **10/10**

By the alternate segment theorem (AB tangent to $\odot BMD$ at B, AC tangent to $\odot CMD$ at C),

$$\angle ABC = \angle BDM, \qquad \angle ACB = \angle CDM.$$

So $\angle BAC = 180^\circ - \angle ABC - \angle ACB = 180^\circ - \angle BDM - \angle CDM = 180^\circ - \angle BDC$, hence $ABDC$ is cyclic.

From $\angle BAD = \angle BCD = \angle MCD$ (same segment) and $\angle MDC = \angle ACB = \angle BDA$, we get $\triangle ABD \sim \triangle CMD$, so

$$\frac{DC}{DA} = \frac{MC}{BA} = \frac{MB}{BA}.$$

Since $\angle ABM = \angle ABC = \angle ADC$ (same segment), $\triangle ABM \sim \triangle ADC$ (SAS, or directly from the spiral similarity at A).

Hence $\angle BAM = \angle DAC = \angle DBC = \angle DBM$. Also $\angle ABM = \angle BDM$ (tangent at B), so $\triangle ABM \sim \triangle BDM$ and

$$\frac{MA}{MB} = \frac{MB}{MD} \implies MA \times MD = MB^2 = MB \times MC.$$

Marker's comment. Complete, with a reason for every angle equality. The configuration (D on the far side of BC, so A, B, D, C in that order on the circle) is read from the diagram, which BMO1 markers accept. One line would settle the point. If D were on the same side as A, then $\angle BDM + \angle CDM = 360^\circ - \angle B - \angle C > 180^\circ$, but this sum is $\angle BDC < 180^\circ$.

Script B **7/10**

We have $\angle ABC = \angle BDM$ and $\angle ACB = \angle CDM$, so $\angle BDC = \angle ABC + \angle ACB = 180^\circ - \angle BAC$ and $ABDC$ is cyclic.

Then $\angle BAD = \angle MCD$ and $\angle BDA = \angle ACB = \angle MDC$, so triangles ABD and CMD are similar and $DC/DA = MC/AB = MB/AB$. With $\angle ABM = \angle ADC$ this gives triangles ABM and ADC similar, so $\angle BAM = \angle DAC = \angle DBM$. Also $\angle ABM = \angle BDM$. So triangles ABM and BDM are similar and $MA \cdot MD = MB^2 = MB \cdot MC$.

Marker's comment. The right chain in the right order, but no angle equality has a reason. The first and last come from the tangents by the alternate segment theorem, and the rest are angles in the same segment of circle $ABDC$. The markers' report penalised exactly this, and naming the theorem at each step gives full marks.

Script C **4/10**

By the alternate segment theorem in the circle through B, M, D, tangent to AB at B, $\angle ABM = \angle BDM$. Similarly $\angle ACM = \angle CDM$. Adding,

$$\angle BDC = \angle BDM + \angle MDC = \angle ABC + \angle ACB = 180^\circ - \angle BAC,$$

so A, B, D, C are concyclic.

Also $\angle MBD = \angle MAB$. So triangles MBD and MAB have $\angle MBD = \angle MAB$ and $\angle BDM = \angle ABM$, so they are similar. Hence $MB/MA = MD/MB$, so $MA \times MD = MB^2 = MB \times MC$.

Marker's comment. The concyclicity is proved properly and earns real credit. The second paragraph rests on $\angle MBD = \angle MAB$, which is true but is the whole difficulty, and no single circle theorem gives the equality since the two angles do not stand on a common chord. Script A needs two similar triangles to prove this step, and the report says such assumed equalities were heavily penalised.

Script D **2/10**

By the alternate segment theorem, $\angle ABC = \angle BDM$ since AB is tangent to circle BMD at B, and $\angle ACB = \angle CDM$ since AC is tangent to circle CMD at C. So $\angle BDC = \angle ABC + \angle ACB = 180^\circ - \angle BAC$ and $ABDC$ is a cyclic quadrilateral.

The chords AD and BC of this circle meet at M, so by the intersecting chords theorem $MA \times MD = MB \times MC$.

Marker's comment. The concyclicity earns the marks. The chords AD and BC meet at E, which is M only when $AB = AC$, so intersecting chords gives $EA \times ED = EB \times EC$ and nothing about M. A scalene diagram with AB clearly longer than AC would have shown the error.

What full marks needs

Prove that $ABDC$ is cyclic, and name the alternate segment theorem for both tangents. Then reach $\angle BAM = \angle DBM$ with reasons, which together with $\angle ABM = \angle BDM$ gives $\triangle ABM \sim \triangle BDM$ and $MA \times MD = MB^2$.